

HeroPips — Functional Requirements Document (FRD v1.0)

Product: HeroPips (heropips.com) **Document type:** Functional Requirements Document (system behavior specification) **Owner:** Product/BA · **Primary consumers:** Engineering, QA, Security **Status:** Draft for approval · **Date:** July 24, 2026

Version	Date	Author	Change
0.1	Jul 24, 2026	Product + Eng lead	First end-to-end draft
1.0	—	—	Approved baseline (pending review)

References: HeroPips BRD v1.0 (business requirements BR-01...BR-26, business rules BRL-01...BRL-11) · HeroPips PRD v1.0 (epics E1-E19, pricing §7, NFRs §10, architecture §11, compliance §14) · Strategy Plan.

1. Introduction & Conventions

1.1 Purpose. This FRD specifies *how the system shall behave* to satisfy the BRD/PRD: detailed functional requirements, field-level validation, state machines, calculations, error handling, and interface behavior for Phase 0 → GA v1.0. Phase 2/3 items are included only where they impose architectural obligations now.

1.2 Requirement convention. IDs are `FR-<module>.<seq>` (e.g., FR-07.03). Language: "shall" = mandatory; "should" = recommended; "may" = optional. Priorities inherit the PRD: **P0** (must-ship in the stated phase), **P1** (fast-follow), **P2** (future/architectural). Acceptance criteria (AC) use Given/When/Then. Each module header lists its BRD/PRD traceability.

1.3 Global definitions. "Live account" = a broker/exchange connection capable of real orders. "Paper account" = simulated account. "Execution pipeline" = ingestion → risk engine → order router → connector (PRD §11). "Tenant config" = the versioned JSON object of E16.2. All timestamps UTC (timestamp_{tz}); all money integer minor units + ISO currency; all IDs UUIDv7 unless provider-native.

1.4 Global functional rules (apply to every module).

- FR-00.01 (P0)** Every state-changing operation shall be authorized server-side against RBAC + entitlements; client-side gating is presentational only. (BR-15, BRL-01)
- FR-00.02 (P0)** Every state-changing operation shall write an audit record {actor, tenant, action, entity, before, after, ip, ts}; audit records are append-only. (BR-23)
- FR-00.03 (P0)** All list endpoints shall be tenant-scoped (tenant_id) and paginated (cursor, default 50, max 200). (BR-24)
- FR-00.04 (P0)** All user-visible errors shall map to a catalog entry (Appendix A) with stable `error_code`, human copy, and remediation hint; raw provider errors are never shown verbatim.
- FR-00.05 (P0)** All idempotent write endpoints shall accept an `Idempotency-Key` header; retried keys return the original result for 24h.

- **FR-00.06 (P0)** Compliance surfaces: any screen rendering signals, tips, backtests, performance, or share assets shall include the standard disclaimer component (version-tracked). (BRL-09)

2. System Overview

2.1 Actors.

Actor	Description	Key permissions
Visitor	Unauthenticated	Marketing pages, signup, delayed signal view (read)
Free user	Registered, no paid plan	Academy core, paper trading, delayed signals, no live connect
Paid user (Starter/Pro/Elite)	Active subscription or LTD grant	Per entitlement matrix (PRD §7.1)
Affiliate partner	Approved partner account	Partner portal only (links, stats, payouts)
Admin roles	SuperAdmin, Ops, Content, Support, Compliance, Finance	Per RBAC matrix (FRD-17.6)
System	Schedulers, workers, reconcilers	Signal evaluation, sync, expiry, payouts prep
External systems	TradingView, MetaAPI, CCXT venues, cTrader, Stripe, SendGrid, calendar feed, PostHog	Via integration contracts (§5)

2.2 Module map (20 modules). 01 Identity & Access · 02 Onboarding · 03 Academy · 04 Paper Trading · 05 Broker Connectivity · 06 Mirroring & Reconciliation · 07 Trade Ticket & Orders · 08 Signal Engine · 09 TradingView Webhooks · 10 Risk Engine (Trade Guard) · 11 Runtime Backtesting · 12 Share Cards & Signal Lists · 13 Points & Gamification · 14 Referral Engine · 15 Affiliate/IB Engine · 16 Billing & Entitlements · 17 Admin & Tenant Config · 18 Notifications · 19 Dashboards & Journal · 20 Multi-Tenancy Foundations.

2.3 Traceability (module ↔ PRD epic ↔ BRD).

FRD module	PRD epic	BRD	Phase
01	E1	BR-01	Phase 0
02	E1.3	BR-01	Phase 0
03	E2	BR-01, BR-03	Phase 0 → GA
04	E3	BR-01, BR-02	Phase 0
05	E4	BR-05, BR-06	Phase 0 → GA
06	E5	BR-10	Phase 0
07	E6	BR-09	Phase 0
08	E7	BR-08	Closed Beta
09	E8	BR-07	Closed Beta
10	E9	BR-11, BR-12, BR-13	Closed Beta → GA
11	E10	BR-14	GA
12	E11	BR-04	Open Beta
13	E12	BR-20	Closed Beta
14	E13	BR-18	Closed Beta
15	E14	BR-19, BR-17	Open Beta
16	E15, §7	BR-15, BR-16	Phase 0 → Open Beta
17	E16	BR-22, BR-23	Phase 0
18	E17	BR-09	Closed Beta
19	E18	BR-09, BR-25	Closed Beta
20	E19	BR-24	Phase 0 (architecture)

3. Detailed Functional Requirements by Module

FRD-01 — Identity & Access Management (E1 · Phase 0)

FR-01.01 (P0) Registration. The system shall register users via email+password or Google OAuth. Field validation:

Field	Rules
email	RFC 5322 format, ≤254 chars, lowercased, unique per tenant; disposable-domain blocklist (configurable)
password	≥10 chars, ≥3 of 4 classes (upper/lower/digit/symbol), not in breached-password list, ≠ email local part
display_name	3–30 chars, unicode letters/digits/space/underscore, profanity filter
country	ISO-3166-1 alpha-2, required (drives compliance flags FRD-17); default from IP geolocation, user-editable
timezone	IANA tz, default from browser
consents	risk acknowledgement + ToS + Privacy checkboxes, each stored as {doc_id, version, ts, ip} — all required (BRL-09 context, PRD E1.2)

FR-01.02 (P0) Email verification. Signup shall send a single-use verification token (SendGrid), 24h expiry; unverified accounts may browse academy previews only; resend limited to 3/hour. Verification fires `signup_completed`.

FR-01.03 (P0) Login & logout. 5 failed attempts within 15 min shall lock the account for 15 min and require CAPTCHA; error copy shall be generic ("email or password incorrect") to prevent enumeration; all attempts logged with IP/device.

FR-01.04 (P0) 2FA. TOTP (RFC 6238, 30s window, ±1 step drift) with QR enrollment and 10 single-use backup codes. The system shall block the live-connection flow (FRD-05) server-side unless 2FA is enabled. Disabling 2FA shall require current TOTP + password and shall notify by email.

FR-01.05 (P0) Sessions. Access JWT 15 min; rotating refresh token 30 days, revoked on password change/2FA change; device list (agent, ip, last_seen) with per-device revoke; max 10 concurrent sessions.

FR-01.06 (P0) Password reset. Single-use token, 60 min expiry; reset revokes all sessions and notifies by email.

FR-01.07 (P1) Profile management. Avatar upload (≤2MB, jpg/png/webp, virus-scanned, resized 256px); country change shall re-evaluate compliance flags immediately and may disable features with an explanatory banner.

FR-01.08 (P2) Phone/OTP login (MENA/India): architecture shall keep the identity provider abstraction pluggable.

AC (FR-01.04): Given a Starter user without 2FA, when they submit the MT5 connect form, then the API returns 403 `error_code=2fa_required` and the UI routes to 2FA setup, resuming the connect flow after enrollment.

FRD-02 — Onboarding & Path Routing (E1.3 • Phase 0)

FR-02.01 (P0) Quiz. After verification, the system shall present a 3-question quiz — experience (none / <1y / 1y+), goal (learn / automate my strategy / trade crypto / pass a prop challenge), market (forex / crypto / stocks / unsure) — skippable at any step.

FR-02.02 (P0) Routing rules.

Answers	Route	Side effects
experience=none OR goal=learn	Academy Lesson 1	Paper account pre-created; <code>onboarding_path_selected{path: learn}</code>
goal=automate	Webhook/signal intro screen → paper target	Paper account pre-created; path=connect_later
goal=trade crypto AND experience≥<1y	Connect CEX explainer (paywall- aware)	path=connect
goal=prop challenge	Trade Guard explainer → Elite pitch	path=guard
skipped	Dashboard with 3- item "Start here" checklist (do a lesson, place a paper trade, explore signals)	path=skipped

FR-02.03 (P1) Checklist widget. Persistent until all 3 items complete or dismissed; completion of all 3 awards points (FRD-13) and fires `activated=true` if within 24h of signup.

FRD-03 — Academy: Content, Quizzes, Certificates (E2 • Phase 0 v0 → GA full)

FR-03.01 (P0) Content model. tracks → modules → lessons. Lesson fields: id, slug, title (≤80), summary (≤200), body (rich text/MDX), media refs (video URL, images), est_minutes, free_flag, points_award (default 50), status (draft/published/retired), version, updated_by. Publishing shall create an immutable version snapshot; learners always see the latest published version.

FR-03.02 (P0) Quiz engine. Per lesson: 1-10 questions, MCQ single- or multi-select, optional per-question explanation shown after submit. Pass mark ≥70%. Attempts unlimited; question order and option order shall randomize per attempt; the best score is stored. Passing marks the lesson complete exactly once and awards points once (FRD-13 dedupe).

FR-03.03 (P0) Progress & gating. Track progress = completed lessons ÷ published lessons. Paid-flagged lessons shall render a preview (first 20%) + paywall for non-entitled users (`paywall_viewed{trigger: lesson}`) per BRL-01/E7.5 semantics.

FR-03.04 (P0) Admin CMS. Content role shall create/edit/reorder/publish/retire lessons and quizzes with draft preview; no deploy required; all changes audit-logged (FR-00.02).

FR-03.05 (P1) Practice deep links. A lesson may define a CTA `{screen:"paper_ticket", prefill:{symbol,type,side,qty}}`; clicking shall open the paper ticket with fields pre-filled and fire `lesson_cta_practice`.

FR-03.06 (P1) Certificates. On track completion, the system shall render a 1200×630 PNG from the tenant-branded template: display name, track, date, verification code (public check URL), and the disclaimer strip; downloadable and shareable via FRD-12 mechanics.

FR-03.07 (P2) Cohorts/live sessions; AR/Hi localization (copy externalized to i18n from day 1).

FRD-04 — Paper Trading Engine (E3 · Phase 0)

FR-04.01 (P0) Accounts. Each user shall get one paper account by default (Pro: 3, Elite: 5 per PRD §7.1): starting balance \$10,000 USD, leverage 1:100 for FX/metals, 1:1 spot crypto. Reset shall require confirmation, archive prior history (viewable, flagged `reset_generation=n`), and restore the starting balance.

FR-04.02 (P0) Price source. Quotes from the market-data service (FRD-11 store) — live where available, else delayed ≤15 min with a persistent "Delayed data" label (Open Question Q9 default). Spread applied = per-symbol configured typical spread (admin table; e.g., EURUSD 1.0 pip default).

FR-04.03 (P0) Fill simulation.

Order type	Fill rule
Market	Fill immediately at current bid (sell) / ask (buy)
Limit	Fill when opposite-side price touches limit price (buy: ask ≤ price)
Stop	Trigger when price touches stop; then fill as market
Stop-limit	Trigger at stop; place limit; fill per limit rule; expire per TIF
SL/TP on positions	Evaluated on every tick/bar update; SL uses worst side, TP uses favorable touch

FR-04.04 (P0) Accounting. PnL (FX): $(close - open) \times direction \times contract_size \times lots \times conversion\text{-}to\text{-}USD$. Crypto spot: $qty \times (close - open)$. Equity = balance + Σ floating PnL. Margin used = $\Sigma (lots \times contract / leverage \times conv)$. Margin level = equity ÷ margin used; at margin level ≤ 50% the engine shall auto-close the largest losing position (stop-out) and notify.

FR-04.05 (P0) Parity. The paper ticket shall be the same component/validation as FRD-07 with `account.kind=paper`; every downstream service (signals, webhooks, guards) shall

accept paper targets; paper orders bypass the broker router and route to the simulator, always flagged (`simulated=true`).

FR-04.06 (P1) Leaderboard. Opt-in monthly leaderboard by % return (min 10 closed trades to rank), display-name only, resets monthly, top-100 visible; excluded on opt-out at any time.

AC (FR-04.04): Given a paper account with equity \$10,000 and one EURUSD 1.0-lot long, when floating loss reaches −\$5,000 and margin level hits 50%, then the engine closes the position at market, records reason=stop_out, and sends the stop-out notification exactly once.

FRD-05 — Broker & Exchange Connectivity (E4 · Phase 0: MT5 · Open Beta: CCXT · GA: cTrader)

FR-05.01 (P0) Connector abstraction. All product code shall call (`IBrokerConnector`) only: (`connect`, `getAccountInfo`, `getPositions`, `getOrders`, `placeOrder`, `modifyOrder`, `cancelOrder`, `closePosition`, `getHistory(from,to,cursor)`, `subscribeStream`, `disconnect`). Adapters: `MetaApiAdapter`, `CcxtAdapter(exchange)`, `CTraderAdapter` (GA). Adding a venue shall require zero changes outside the adapter + venue config.

FR-05.02 (P0) Entitlement gate. Creating a live connection shall check (`entitlements.live_connections_max`) server-side; Free = 0 → 402 (`error_code=upgrade_required`) with plan sheet payload (BRL-01). The connect UI is browsable by Free users up to credential entry (conversion point, (`paywall_viewed{trigger:connect}`)).

FR-05.03 (P0) MT5 connect wizard (MetaAPI). Fields & validation:

Field	Rules
account_number	digits, 5-12
password	masked; never logged; investor (read-only) passwords shall be detected via a post-provision trade-capability probe and rejected with (<code>error_code=investor_password</code>)
server	autocomplete from MetaAPI server directory; free-text fallback with "server not found" guidance
nickname	3-30 chars, unique per user

Provisioning shall map MetaAPI states → HeroPips lifecycle: created→deploying = (`connecting`); deployed/connecting = (`synchronizing`); connected+synchronized = (`live`); broken stream = (`degraded`); undeployed/failed = (`disconnected`). Timeout: if not (`live`) within 3 min, show retry + support link; keep provisioning in background up to 15 min.

FR-05.04 (P0) MT5 error mapping (excerpt — full list Appendix A). invalid credentials → "Check login/password (use the *trading* password, not investor)"; server not found →

suggest exact server string from broker email; broker restriction/EA-disabled → guidance article link.

FR-05.05 (P0) CEX connect (CCXT: Binance, Bybit). Fields: exchange (enum), api_key, api_secret, passphrase (if venue requires). Validation sequence: (1) `fetchBalance` proves read; (2) permission probe — Binance `sapi/v1/account/apiRestrictions` (or equivalent), Bybit key-info — **if withdrawal is enabled the connection shall be rejected with remediation copy and nothing stored** (BRL-05); (3) a canary `fetchOpenOrders` proves trade scope readability. IP-allowlist instructions (our egress IPs) shall be displayed pre-save.

FR-05.06 (P0) Secret handling. Secrets shall be envelope-encrypted (AES-256-GCM data key wrapped by cloud KMS master key), stored as `encrypted_secret_ref`; decrypt only in the connector process memory; never in logs/traces (log scrubber test in CI); "reveal" is not offered; delete shall undeploy/revoke at the provider, purge the secret, and retain trade history flagged `connection_deleted`.

FR-05.07 (P0) Status surfacing. A status chip (live=green, synchronizing=blue, degraded=amber, disconnected=red) shall render on every screen that can trigger a trading action; degraded/disconnected shall disable order submission with reason.

FR-05.08 (P1) Health page. Per connection: provider, region, last sync ts, stream latency (ms rolling p50/p95), last 20 errors, reconnect button, "send test order (paper route)".

FR-05.09 (P1, GA) cTrader. OAuth2 (accounts + trading scopes), token refresh, account read + market/limit orders v1; same lifecycle states.

AC (FR-05.05): Given a Binance key with withdrawal permission, when validation runs, then the API returns 422 `error_code=key_scope_withdrawal`, the UI shows the exact remediation ("create a key with only Reading + Spot Trading"), and no record or secret is persisted.

FRD-06 — Mirroring, Streaming & Reconciliation (E5 · Phase 0)

FR-06.01 (P0) Initial sync. On `live`, the system shall import in order: account info → open positions → working orders → closed-deal history (paged, up to 5 years where the provider allows), with a progress indicator; partial-history providers shall record `history_from` watermark.

FR-06.02 (P0) Streaming upserts. Provider stream events (order update, position update, deal added, balance/equity) shall upsert local records keyed by provider-native IDs within ≤3s end-to-end to UI; unknown IDs trigger a targeted fetch.

FR-06.03 (P0) Reconciler. Every 5 min per live account: fetch snapshot; diff positions/orders by provider ID and balance within ±0.01; auto-correct local to provider truth; write `sync_anomaly {type: missing_local|missing_remote|field_mismatch, entity, before, after}`; externally-originated changes attributed `source=external`. ≥3 anomalies/hour on one account shall page ops (FRD-18 ops alert).

FR-06.04 (P0) Persistence. Tables per §6 data dictionary; provider IDs unique-indexed per connection for idempotent upserts.

FR-06.05 (P1) Export. CSV per account (history, positions) with documented columns; async job + download link (24h expiry).

FR-06.06 (P1) Portfolio rollup (Pro/Elite). Aggregate equity/PnL across connections with per-currency conversion at daily reference rates; base currency user-selectable (USD default).

FRD-07 — Trade Ticket & Order Management (E6 · Phase 0)

FR-07.01 (P0) Ticket fields & validation.

Field	Rules
account	active, status=live (or paper); entitlement-checked
symbol	must exist in venue symbol cache (refreshed 24h); shows contract size, min/step volume, stops level
side	buy/sell
type	market / limit / stop / stop_limit
quantity	mode lots (FX: min/step per symbol spec, e.g., 0.01) or units (crypto: venue min notional & step) — validated against venue limits
price / stop_price	required per type; tick-size rounded; limit buy < market context warnings
sl / tp	absolute price, or relative (pips / %); must respect venue stops-level distance; SL on the loss side, TP on the profit side (validated per direction)
tif	GTC (default) / DAY / IOC where venue supports; unsupported options hidden
comment	≤31 chars (MT5 constraint), sanitized

FR-07.02 (P0) Live calculations (client + server authoritative). Pip/point value = $\text{contract_size} \times \text{point} \times \text{conversion}$; est. margin = $\text{lots} \times \text{contract} \div \text{leverage} \times \text{conversion}$; risk \$ = $|\text{entry} - \text{SL}| \text{ in points} \times \text{point value} \times \text{lots}$. **Risk-% sizing:** $\text{lots} = \text{floor_to_step}(\text{equity} \times \text{risk_pct} \div (\text{SL_distance_points} \times \text{point_value_per_lot}))$; if result < venue min volume → block with `error_code=risk_too_small_for_min_lot`; clamp to symbol max and guard caps (FRD-10).

FR-07.03 (P0) Order state machine. created → validated → queued → routed → accepted → partially_filled → filled | rejected | cancelled | expired. Each transition shall record ts, actor/source (manual|signal|webhook|external|system), and provider payload snapshot; illegal transitions shall be impossible at the persistence layer (state check constraint).

FR-07.04 (P0) Position operations. Close full/partial (partial $\geq$ venue min step), reverse (close + opposite market order as one intent, two provider ops, compensating rollback on second-leg failure = position remains closed and user notified), modify SL/TP (stops-level validated), cancel/modify working orders.

FR-07.05 (P0) Provider error mapping. A mapping table shall translate provider errors → catalog entries (Appendix A). Excerpt: MT5 10019/no-money → "Not enough margin — reduce size"; 10016 invalid stops → "SL/TP too close: min distance X points"; 10018 market closed; CCXT InsufficientFunds / InvalidOrder / RateLimitExceeded → mapped copy + retry-after where applicable. Unmapped errors show a generic message + support code and are logged for triage.

FR-07.06 (P1) Panic Close. Button per account: modal requires typing CLOSE; then cancel all working orders, then close all positions sequentially (largest exposure first); a results report lists per-item success/failure; partial failures leave a persistent task banner with retry. Free tier: paper accounts only.

FR-07.07 (P1) Quantity helpers. Fixed lots / % of equity / risk-% (FR-07.02) selectable and remembered per account.

AC (FR-07.03): Given a routed order rejected by the broker, when the rejection event arrives, then local state = rejected with mapped reason, no phantom position exists, and the user receives exactly one rejection notification.

FRD-08 — Signal Engine: Library, Builder, Modes, AI Tips (E7 · Closed Beta)

FR-08.01 (P0) Predefined strategy definition (admin-managed). Fields: id, version, name, human-readable rule_text, param schema (JSON Schema with defaults + bounds), default risk block, allowed markets/timeframes, tier availability, hypothetical stats {dataset_version, period, winrate, pf, maxdd} rendered with the disclaimer component. Retiring a strategy shall pause dependent instances with notification, never delete them.

FR-08.02 (P0) Instance configuration. Schema per PRD Appendix B. Validation bounds: risk_pct 0.1-2.0 (admin-raisable per tenant), max_concurrent 1-10, symbols count $\leq$ tier limit, session windows non-overlapping valid ranges, target_account owned + entitled. Instances are versioned; edits create version n+1; running evaluations always use the pinned version.

FR-08.03 (P0) Custom builder DSL. Condition tree: $\{\text{all: [...]}\} \mid \{\text{any: [...]}\}$ of predicates $\{\text{ind, params, op, ref}\}$. v1 indicators: EMA, SMA, RSI, MACD, ATR, PRICE, LEVEL. Ops: crosses_above, crosses_below, gt, lt, gte, lte, within_pct. Limits: depth ≤ 3 , total predicates ≤ 12 . Server shall validate the tree, dry-run it on the last 100 bars, and show fired-bar preview before enabling.

FR-08.04 (P0) Evaluation scheduler. Per timeframe, at bar close +2s grace, the evaluator shall process enabled instances against the latest closed candle from the data store; evaluation is skipped (and logged) when the market/session is closed or data is stale >2 bars. Dedupe key = (instance_id, symbol, tf, bar_ts) — re-emissions (restart, replay) shall be dropped with $\text{signal_duplicate_dropped}$.

FR-08.05 (P0) Execution modes.

Mode	Behavior
notify	Notification only (FRD-18), no order intent
confirm	Single-use signed action token, 15-min expiry; clicking executes through the pipeline; expired tokens show "signal expired"
auto	Emits order_intent directly; enabling requires: versioned consent record + active drawdown guard on target account (BRL-04); consent re-prompt every 90 days via job; missing prerequisites block enable with inline fix-flow

FR-08.06 (P0) Order intent emission. `(order_intent {tenant, user, account, source:signal, instance_id, version, symbol, side, template(type, qty|risk_pct, sl, tp), idempotency_key=hash(instance,symbol,tf,bar_ts)})` → execution pipeline (always through FRD-10).

FR-08.07 (P0) AI trade tips. On ticket open and on fired signals, the tips service shall produce ≤3 educational observations from deterministic rules first (SL distance vs ATR(14) outside 0.5–3.0×; risk_pct > user norm; session/news proximity via FRD-10 data; correlated exposure e.g. EURUSD+GBPUSD same direction), optionally phrased via LLM with a locked template. Tips shall never contain directives ("buy", "sell", "you should"); a lint list blocks output containing directive verbs; every tip renders the disclaimer microcopy + related lesson link. Tips are logged per trade for E18.2 display. (BR-08, Open Question Q4 governs copy.)

FR-08.08 (P1) Instance dashboard. Status, last evaluation ts/result, fired history (30d), per-instance realized PnL attribution (orders with source=signal & instance_id), pause/resume.

FR-08.09 (P1) Free delayed view. Free tier sees library signals fire on a 15-min delayed feed, watch-only, with upgrade CTA; no execution affordances render.

AC (FR-08.05-auto): Given an instance switched to auto without a drawdown guard on the target account, when the user clicks Enable, then the API returns 409 `(error_code=guard_required)`, and the UI opens the guard-creation flow pre-scoped to that account.

FRD-09 — TradingView Webhook Automation (E8 · Closed Beta)

FR-09.01 (P0) Endpoint management. Users shall create endpoints: `(https://hooks.heropips.com/wh/{uuid})`; per-endpoint 32-byte secret displayed once (regenerate invalidates old); target account (paper/live), default order template (type, qty-mode + value, default SL/TP rule), status enabled/paused. Copy-paste TradingView alert message templates shall be generated per endpoint.

FR-09.02 (P0) Payload schema v1 (JSON, ≤8KB).

Field	Req	Rules
key	conditional	endpoint secret (TLS-only fallback auth when HMAC header absent — TradingView limitation)
idempotency_key	optional	≤128 chars; recommended template <code>tv-{{time}}-{{ticker}}-{{strategy.order.action}}</code>
action	required	buy / sell / close / close_all (P1: close_partial{pct}, move_sl, flatten_symbol)
symbol	required for non-close_all	must resolve via endpoint symbol map
order_type	optional	market (default) / limit / stop
qty XOR risk_pct	optional	else endpoint default template applies; bounds validated per FRD-07
price, sl, tp	optional	numeric; validated per FRD-07 rules
timestamp	required	unix seconds; reject if
comment	optional	≤31 chars

FR-09.03 (P0) Authentication & anti-replay. Order of checks: (1) HTTPS only; (2) HMAC-SHA256 of raw body vs `X-HP-Signature` (constant-time compare) — else `key` field match; (3) timestamp window ±300s; (4) nonce/fingerprint cache 10 min. Failures → 401, one console entry, counter++; ≥20 failures/5min from one IP → temporary IP block (Cloudflare rule) + user notified once.

FR-09.04 (P0) Idempotency. Explicit key, else fingerprint = hash(endpoint, action, symbol, qty|risk, price, 10s bucket). Keys stored 24h; duplicates return 200 `{status:"duplicate", original_order_id}` and increment `webhook_deduped`.

FR-09.05 (P0) Budgets. Monthly alert counters per tier (100/1,000/10,000); at 80% send one email; over budget → 429 `{status:"budget_exceeded"}` + at most one email/day; counter resets on billing anchor. Paused/disabled endpoints → 410.

FR-09.06 (P0) Symbol mapping. Per-endpoint map `tv_ticker → venue_symbol` with auto-suggestions (normalized matching, e.g., OANDA:EURUSD → EURUSD.a); unmapped symbol → 422 `unmapped_symbol`, console entry with one-click "add mapping" pre-filled.

FR-09.07 (P0) Processing pipeline (synchronous, target p50 ≤1.5s). verify → parse/validate → map → build order_intent (source=webhook) → risk engine (FRD-10) → route → respond `{status: executed|blocked_by_guard|duplicate|error, order_id?, reason_code?}`. Guard blocks return HTTP 200 with status=blocked_by_guard (TradingView treats non-2xx as failure/retry) + user notification.

FR-09.08 (P1) Console & test-fire. Last 30 days (90 for Elite) of received payloads: raw (secrets masked), parse result, decision, broker response, latency; test-fire tool sends a sample payload flagged test=true which always routes to the paper simulator regardless of target.

AC (FR-09.04): Given the same alert delivered 3× in 8s, when processing completes, then exactly one broker order exists, responses are executed + duplicate + duplicate, and the console shows 1 executed + 2 deduplicated.

FRD-10 — Risk Engine "Trade Guard" (E9 · Closed Beta v1 → GA full)

FR-10.01 (P0) Guard model. guards {id, account_id, type, params(JSON, schema-validated), status active/paused, created_by, challenge_tag?}. Guards are account-scoped; evaluation is server-side and synchronous in the execution path for every order intent (manual, signal, webhook).

FR-10.02 (P0) Evaluation pipeline & precedence (fixed order). 1 account flags (panic in progress, connection degraded) → 2 compliance/country flags (FRD-17) → 3 circuit breakers (daily-loss, max-drawdown state) → 4 exposure caps (max concurrent, per-symbol/currency) → 5 session filter → 6 news filter → 7 sizing transform (risk-% → volume; FR-07.02) → 8 per-trade risk cap → 9 slippage transform (market → marketable-limit at max deviation where supported). Output: `allow(final_order)` (possibly transformed) or `block(reason_code)`. First block wins; risk-reducing actions (close, reduce, cancel, SL tighten) shall bypass entry-blocking guards 3-6.

FR-10.03 (P0) Fail-closed. If guard evaluation errors or exceeds 800ms for a **live** intent, the intent shall be blocked with `system_guard_unavailable`, queued visible to the user ("automation paused — safety first"), and ops alerted; paper intents may proceed flagged `guard_skipped=true`. (BRL-07)

FR-10.04 (P0) Guard specifications (params · logic · block code).

Guard	Params (bounds)	Logic	Code
Risk-% sizing	risk_pct 0.1–5.0	transform per FR-07.02	(transform)
Per-trade risk cap	max_risk_pct / max_risk_amt	computed risk ≤ cap else block	risk_cap
Max concurrent	n 1–50	open positions on account < n	max_concurrent
Max exposure/symbol	lots or notional per symbol & per currency leg	Σ open + pending ≤ cap	exposure_cap
Daily-loss breaker	threshold_pct 0.5–20 or amount; action block flatten_and_block; day boundary = broker server day (fallback account tz)	day realized (closed deals today) + floating ≤ –threshold → trigger; entries blocked until next day; hard variant closes all first	daily_loss
Max-drawdown watchdog	dd_pct 1–50 from peak equity (high-water since guard creation); action alert block flatten	equity ≤ peak × (1 – dd) → action; block persists until user acknowledges + edits guard	max_drawdown
Session filter	windows[{tz, start, end}] ≤ 4	entries outside windows blocked	session_filter
News filter (P1)	impact=high; window ±5–60 min; currency match on symbol legs	entries within window of matching event blocked	news_filter
Slippage cap (P1)	max_deviation points	market → marketable limit @ quote±deviation where supported; else post-fill check:	fill – quote
Break-even (P1)	trigger_rr 0.5–5 or pips; offset points	once profit ≥ trigger: SL → entry + offset, once, idempotent	(post-trade)
Trailing stop (P1)	fixed pips or ATR(period 5– 50)×mult 0.5–5; update throttle 5s	SL ratchets favorably only	(post-trade)
Partial TP ladder (P1)	≤3 levels {at_rr, close_pct} Σ ≤100%	close pct at each RR level, once per level	(post-trade)

Guard	Params (bounds)	Logic	Code
Stale-order cancel (P1)	age > N min/hours or price drift > X%	cancel + notify	stale_cancel

FR-10.05 (P0) Notification collapse. Repeated blocks of the same code on one account shall notify once, then collapse into a 10-minute summary (FRD-18.4).

FR-10.06 (P1, GA) Prop-firm presets (Elite). Generic templates ("1-step", "2-step" challenge: daily-loss %, max-loss %, min trading days counter) bundling guards above; while `challenge_tag` is active, loosening any bundled param shall require a 24h cooldown with explicit warning. Firm names are not used pending Open Question Q6.

AC (FR-10.02/daily_loss): Given a 3% daily breaker and day loss at -3.1%, when a webhook buy intent arrives, then the response is `blocked_by_guard:daily_loss`, exactly one notification is sent for the storm, and a close-position request on the same account succeeds.

FRD-11 — Runtime Backtesting & Market Data (E10 · GA)

FR-11.01 (P0) Data store. TimescaleDB hypertable `ohlcv(symbol, tf, ts, o, h, l, c, v)` with continuous aggregates (M1 → M5/M15/H1/H4/D1); launch set: 28 FX pairs + gold/silver + top-30 crypto; depth 1-15 years per source (Dukascopy FX, Binance dumps); nightly ingestion jobs with gap detection + backfill; `dataset_versions {id, source, range, checksum, created}` — every backtest pins one.

FR-11.02 (P0) Backtest request (Pro+). Inputs: config snapshot (instance version), 1-3 symbols, tf, range ≤5y, initial balance (default \$10,000), assumptions {spread: per-symbol default, fee: venue default, slippage: fixed points (default 1)} — all displayed in results. Entitlement-checked; quotas Pro 20/day, Elite 100/day; concurrency 2/user.

FR-11.03 (P0) Job lifecycle. `queued` → `running(progress %)` → `done` | `failed` | `cancelled`; async workers on read replicas; cancel supported; results retained 90 days (Elite: 1 year).

FR-11.04 (P0) Engine parity & determinism. The backtest evaluator shall be the same shared library as the live evaluator (FRD-08.04), bar-close semantics, no lookahead (indicators computed on closed bars only). Identical (config version, range, dataset_version, assumptions) shall produce byte-identical results.

FR-11.05 (P0) Outputs. Metrics: total return %, win rate, profit factor = gross profit ÷ gross loss, expectancy = (WR×avgWin) – (LR×avgLoss), max drawdown (peak-to-trough on equity curve), trade count, avg R; equity-curve series; trade list {ts_in/out, side, entry/exit, sl/tp, pnl, r_multiple}. All surfaces labeled "Hypothetical — not a recommendation" with assumptions block (FR-00.06).

AC (FR-11.04): Given the same inputs re-run twice, when both jobs finish, then result payload hashes are identical and both display the same dataset_version.

FRD-12 — ROI Share Cards & Curated Signal Lists (E11 · Open Beta)

FR-12.01 (P0) Card generation. Types: paper monthly result, streak, certificate, challenge-preset survival, opt-in live monthly result. Render: 1200×630 PNG via worker <2s from tenant-branded template; contents strictly system-computed (display name, % return — never \$ by default, period, account-kind label "Paper account"/"Live (verified)", level badge, disclaimer strip, referral deep link + UTM). Manual stat editing shall not exist; opt-in per card for live data.

FR-12.02 (P0) Sharing. One-tap share targets (X/IG/TikTok/Telegram/WhatsApp) + copy link; landing page renders the card with signup CTA; events `share_card_created/clicked`).

FR-12.03 (P1) Curated lists (follow v1). Admin-curated lists of predefined-instance templates; "Follow" clones the config into the user's own instance (their accounts, their risk) — no linkage to another user's account (self-directed, BR-08); tier limits per PRD §7.1.

FR-12.04 (P2 architecture) Publisher marketplace entities (publisher profile, subscription, 20% take) shall reuse FRD-15 ledger + KYC; no build in v1.

FRD-13 — Points & Gamification "Hero Points" (E12 · Closed Beta)

FR-13.01 (P0) Earning rules (admin-config table; defaults, per-action dedupe).

Action	Points	Cap
Lesson quiz passed	50	once per lesson
Daily learning/trading streak	10/day	7 days counted per week
First paper trade	100	once
First live connection	200	once
First guarded live trade	150	once
Referral converts to paid	500	≤10 rewarded referrals/mo
Track completed	300	once per track

FR-13.02 (P0) Non-cash constraint (BRL-02). Points shall have no cash-out, transfer, or purchase path; the redemption catalog contains only: subscription time (e.g., 3,000 pts = 1 month Starter — admin-config), feature trials, academy modules, badges, merch vouchers fulfilled by ops. A negative test in CI shall assert no withdraw/transfer endpoint exists.

FR-13.03 (P0) Ledger. Append-only `(points_tx {user, type: earn|redeem|expire|adjust, amount, reason_code, ref, ts})`; balance = Σ; admin adjust requires reason + audit (FR-00.02); redemptions atomically decrement + provision the reward.

FR-13.04 (P1) Levels. Zero→Hero ranks from lifetime earned points (thresholds admin-config); rank renders on profile + share cards.

FR-13.05 (P1) Anti-abuse. Velocity caps per action/day; same-device multi-account detection (fingerprint) flags for review; fraudulent points reversible via adjust with audit; expiry: 12 months of account inactivity → expire job with 30-day prior email.

FRD-14 — Referral Engine (E13 · Closed Beta)

FR-14.01 (P0) Links & attribution. Every user gets code (8-char, unambiguous alphabet) + link. Attribution algorithm: (1) click sets 30-day cookie (last-click wins); (2) code entered at signup overrides cookie; (3) at signup, create `referral_edge{referrer, referee, level:1}`; L2 edge = referrer's referrer computed at signup time; self-referral (same user/payment instrument/device fingerprint) blocked.

FR-14.02 (P0) Reward trigger. Rewards issue only on the referee's **first successful payment**, after a 14-day refund hold: referee → 1 month Starter credit; L1 referrer → 1 month subscription credit (or equivalent points); L2 → reduced value (default 500 points). All rewards post to the unified ledger (FRD-13/15); refund/chargeback within 30 days claws back automatically. (BRL-03)

FR-14.03 (P0) Config schema & compliance gate. `referral_config {levels:[{level, reward:{type: credit_months|points, value}}], max_levels, cash_allowed:false, cookie_days, hold_days}` — tenant-config, versioned. Saving `max_levels > 2` or any `cash` reward type shall require an actor holding the Compliance role AND `legal_signoff:{doc_ref, ts}`; otherwise the save is rejected 403 and the attempt audit-logged. UI shall display the counsel-approved warning text on that screen.

FR-14.04 (P1) Dashboard. Clicks, signups, conversions, pending (in hold), earned; share assets deep-linking FRD-12 cards.

FR-14.05 (P1) Fraud. Same-payment-instrument detection across referrer/referee; conversion velocity anomaly flag → holds pause pending review.

AC (FR-14.03): Given an Ops-role admin sets max_levels=5 with cash rewards, when they save, then the API returns 403 compliance_signoff_required, no config change occurs, and an audit entry records the attempt.

FRD-15 — Affiliate / IB Engine (E14 · Open Beta v1 → GA)

FR-15.01 (P0) Partner lifecycle. apply (form: name, channels, audience, geo, payout prefs) → admin review → approved (portal access at partners.heropips.com) | rejected. KYC (identity doc + payout details) required before first payout, not before promotion.

FR-15.02 (P0) Deal configuration. `affiliate_deal` versioned JSON, schema-validated, effective-dated:

```
json
```

```
{
  "rev_share": {"pct": 25, "months": 12},
  "cpa": {"amount_minor": 5000, "currency": "USD", "event": "first_payment"},
  "ib_rebate": {"share_pct": 60, "tiers": [
    {"upto_lots": 100, "per_lot_minor": 400},
    {"upto_lots": 500, "per_lot_minor": 550},
    {"upto_lots": null, "per_lot_minor": 700}]}],
  "sub": {"enabled": true, "max_depth": 2, "master_split_pct": 30},
  "cookie_days": 60, "payout": {"min_minor": 5000, "cycle": "monthly"}
}
```

Templates (Standard / Plus / Custom) + per-affiliate overrides; any field combination valid if schema passes; `sub.max_depth` hard cap 3 (BR-19). Activating a new version shall not retro-change already-accrued entries.

FR-15.03 (P0) Attribution. Dedicated links/coupons; precedence: an explicit affiliate coupon/link wins over consumer referral; a user has exactly one revenue-attribution owner, fixed at first payment. Conversion events consumed from FRD-16 webhooks; broker volume ingested via CSV v1 (columns: broker, account_ref, date, symbol, lots, rebate_amount, broker_deal_id — dedupe on broker_deal_id) → P1 broker API pulls.

FR-15.04 (P0) Double-entry ledger. Every accrual/adjustment/clawback/payout = balanced journal entry in `ledger_entries {journal_id, account (e.g., commission_expense, affiliate_payable:{id}, cash, clawback), direction, amount_minor, currency, ref}`. Invariants enforced by transaction: $\Sigma \text{ debits} = \Sigma \text{ credits}$ per journal; partner payable = accruals – clawbacks – payouts. A nightly balance check failing shall **block the payout queue** and page ops. Sub-IB splits post as two payable legs per the deal split.

FR-15.05 (P1) Payouts. Monthly cycle; eligibility = payable $\geq$ \$50 AND KYC complete; queue → Finance approval (4-eyes for >\$1,000) → executed via bank/Wise/USDT (ops v1) → marked paid with reference; auto-invoice PDF emailed; geo/withholding blocks configurable; failed payout returns to queue with reason.

FR-15.06 (P1) Partner dashboard. Real-time clicks/signups/conversions/volume/accrued/pending/paid (≤ 15 min lag); asset library (banners, copy) served from tenant config.

AC (FR-15.04): Given any date, when the balance job sums a partner's ledger legs, then debits equal credits and payable equals accruals – clawbacks – payouts exactly; any mismatch sets payouts=frozen and alerts ops within 5 minutes.

FRD-16 — Billing, Plans & Entitlements (E15 + §7 · Phase 0 skeleton → Open Beta live)

FR-16.01 (P0) Plan catalog. Tenant-config JSON: `plans[{plan_id, name, price_minor{monthly, annual}, limits{live_connections, paper_accounts, active_signals, custom_configs, webhook_alerts_mo, backtests_day, follow_lists}, flags{ai_tips,`

premium_guards, prop_presets, backtest, priority_queue}}]; hot-reload ≤60s; Stripe Products/Prices synced by a reconciliation job (catalog is source of truth).

FR-16.02 (P0) Checkout & lifecycle. Stripe Checkout for subs (monthly/annual) and one-time SKUs (Academy, LTD); handled webhook events (idempotent by event id): checkout.session.completed, invoice.paid, invoice.payment_failed, customer.subscription.updated, customer.subscription.deleted, charge.refunded. Payment events shall update entitlements within 60s.

FR-16.03 (P0) Entitlement resolution. $\text{effective_entitlements}(\text{user}) = \text{base}(\text{plan}) \oplus \text{grants}(\text{LTD}, \text{promo}, \text{admin})$ evaluated server-side, cached 60s, invalidated on billing events; every gated action calls this service (FR-00.01).

FR-16.04 (P0) Dunning & downgrade. invoice.payment_failed → grace 7 days (banner + emails day 0/3/6) → downgrade to Free: per PRD E7.3, excess live connections become read-only (orders blocked, data visible), excess instances paused, nothing deleted; restoring payment reactivates within 60s.

FR-16.05 (P0) Lifetime Deal. One-time SKU grants `(pro_lifetime)` (Pro limits, 3 live connections, fair-use flags); global seat counter decremented atomically (`(UPDATE ... WHERE seats_remaining > 0 RETURNING)`) — sold-out shows waitlist; cap 500 immutable in code + config checksum (BRL-06); LTD purchases include Academy.

FR-16.06 (P1) Self-serve changes. Upgrade immediate with proration preview; downgrade at period end with impact summary (what pauses/locks); invoices & history page; 14-day money-back on first subscription auto-refund path triggering FRD-14/15 clawbacks.

FR-16.07 (P2) Regional PSP fallback + USDT rail per Open Question Q3; price-book per country.

AC (FR-16.05): Given 1 seat remaining, when two checkouts complete concurrently, then exactly one grant is provisioned, the second is auto-refunded with apology email, and seats_remaining = 0.

FRD-17 — Admin Console & Tenant Configuration (E16 · Phase 0 → continuous)

FR-17.01 (P0) Screens. Users · Plans & entitlements · Signal library · Academy CMS · Referral config · Affiliate deals & payouts · Compliance flags · Tenant/branding · LTD seats · Promos/banners · Ops health · Audit log.

FR-17.02 (P0) Tenant config object (versioned, DB-backed JSON). `({branding:{name, logo_url, favicon, colors{primary, accent, ...}, email_from}, domain, features:{...bool}, plan_catalog_ref, referral_config_ref, affiliate_defaults_ref, compliance:{country_flags:{US:{signals_realtime:false, auto_trade:false, signup:true}, ...}}, disclaimers:{version, texts}, email_templates_ref})`. Publishing creates version n+1 with diff view + one-click rollback; cache-bust propagates ≤60s (AC E16.2).

FR-17.03 (P0) Compliance flags. Country×feature matrix evaluated at session start and on country change; blocked features hide execution affordances and show the counsel-

approved notice; flag changes audit-logged and effective ≤60s. Launch default per Open Question Q5 (US: signals real-time + auto-trade off).

FR-17.04 (P0) User management. Search (email/id/name), 360° view (entitlements, connections, guards, instances, ledgers, consents), suspend/unsuspend (suspend pauses automations + blocks live orders), GDPR export (JSON+CSV bundle ≤72h) and delete (anonymize PII; retain financial/audit records per legal holds).

FR-17.05 (P0) Impersonation. Read-only impersonation with persistent red banner; every viewed screen audit-logged; no write actions possible in impersonation context (server-enforced).

FR-17.06 (P0) RBAC matrix (excerpt).

Capability	SuperAdmin	Ops	Content	Support	Compliance	Finance
Tenant config publish	✓	—	—	—	—	—
Plans/prices edit	✓	—	—	—	—	✓
Signal library CRUD	✓	✓	—	—	—	—
Academy CMS	✓	—	✓	—	—	—
Referral depth/cash change	✓ *	—	—	—	✓ * (signoff)	—
Compliance flags	✓	—	—	—	✓	—
User suspend / GDPR	✓	✓	—	✓ (suspend only)	—	—
Affiliate deals	✓	✓	—	—	—	✓
Payout approval	—	—	—	—	—	✓ (4-eyes >\$1k)
Audit log read	✓	✓	—	—	✓	✓

*Both required for BRL-03 exceptions (FR-14.03).

FR-17.07 (P1) Ops dashboards. Connection fleet (by state), webhook error rate, execution latency p50/p95, MetaAPI spend per account (BO-8 telemetry), guard-block stats, LTD seats, queue depths.

FRD-18 — Notifications & Communications (E17 • Closed Beta)

FR-18.01 (P0) Event→channel matrix (excerpt; templates tenant-branded via SendGrid dynamic templates).

Event	Email	In-app	Telegram (P1)
Verification / security (2FA, new device, password)	✓ (forced)	✓	—
Connection succeeded / degraded / disconnected	✓	✓	✓
Order filled / rejected	pref	✓	✓
Guard triggered / blocked (collapsed)	✓	✓	✓
Signal fired — notify mode / confirm link	✓	✓	✓
Webhook budget 80% / exceeded (1/day)	✓	✓	—
Billing (receipt, dunning, downgrade)	✓ (forced)	✓	—
Payout status (partners)	✓	portal	—

FR-18.02 (P0) Preferences. Per-category toggles (execution, signals, guards, learning, marketing); transactional/security categories cannot be disabled; marketing honors unsubscribe (list-unsubscribe header).

FR-18.03 (P0) Delivery. Outbox pattern: domain events → notifier worker → channel adapters; retries with backoff; SendGrid bounce/complaint webhooks mark addresses and suppress.

FR-18.04 (P0) Batching. Same event-type storms per account collapse to one summary per 10 min (e.g., "17 entries blocked by daily-loss guard since 14:02"); confirm-mode links are never batched.

FR-18.05 (P1) Telegram. Link via deep-link code to the HeroPips bot; execution + guard alerts; unlink from profile.

FRD-19 — Dashboards, Reporting & Journal (E18 · Closed Beta)

FR-19.01 (P0) Home dashboard. Per-account cards: balance, equity, day/week/month PnL, open positions count/exposure, working orders, guard status chips, recent automation feed (last 20 events), connection status.

FR-19.02 (P0) Trade history. Filters: account, symbol, date range, source (manual/signal/webhook/external), status; row → trade detail: event timeline (intent → guard results → route → fills), applied guard outcomes, tips shown (FR-08.07), linked journal entry.

FR-19.03 (P1) Auto-journal. On position close: entry auto-created {trade ref, source, setup tag (from signal/webhook comment), $R \text{ multiple} = \text{pnl} \div \text{initial risk}$ (risk from SL at entry; null-safe), notes (user), screenshot attach ≤3}; weekly summary email (top stats + biggest win/loss + guard saves).

FR-19.04 (P1) Analytics. Win rate, profit factor, expectancy, max DD, avg R, performance by symbol/hour/weekday/strategy/source; paper-vs-live comparison for matched strategies;

formulas identical to FRD-11.05 for consistency.

FR-19.05 (P2) Public opt-in profile with verified (account-linked, unedited) stats — feeds marketplace.

FRD-20 — Multi-Tenancy Foundations (E19 • Phase 0 architecture)

FR-20.01 (P0) Every domain table shall carry `tenant_id NOT NULL`; PostgreSQL RLS policy `tenant_id = current_setting('app.tenant_id')::uuid` on all domain tables; middleware sets the setting from the request host/tenant resolution; a cross-tenant read attempt in tests shall return zero rows.

FR-20.02 (P0) All branding/copy/plan/feature reads shall flow through tenant config (FR-17.02); theming via CSS variables generated from config; CI shall grep-gate hard-coded brand strings ("HeroPips" allowed only in tenant #1 seed data and legal docs).

FR-20.03 (P2) Tenant provisioning (domain + SSL automation, per-tenant SendGrid sender, per-tenant Stripe Connect, isolated affiliate program, tenant-admin role) — Phase 3; schema shall not require migration to enable it.

4. External Interface Requirements

Integration	Purpose	Auth	Key operations / events consumed	Failure handling
MetaAPI (MT5)	Account provisioning, trading, streaming, history	API token (server-side)	provision/deploy account, account info, positions/orders CRUD, deals history, websocket stream	Timeouts + retry w/ jitter; circuit breaker per account; degraded state (FR-05.03); per-account cost telemetry (BO-8)
CCXT (Binance, Bybit v1)	CEX balances, orders, trades, streams	User API keys (encrypted, FR-05.06)	fetchBalance, create/cancel order, fetchOpenOrders, fetchMyTrades, watch* streams	Venue rate-limit aware (token bucket per key); mapped errors (FR-07.05); reject withdrawal-scope keys (BRL-05)
cTrader Open API (GA)	Account read + basic execution	OAuth2 + refresh	accounts, symbols, market/limit orders, positions	Token refresh job; revoke on disconnect
TradingView (inbound)	Alert webhooks	Endpoint secret / HMAC (FR-09.03)	POST alert payloads	200-with-status semantics (FR-09.07); IP blocks via Cloudflare
Stripe	Subscriptions, one-time SKUs, refunds	API key + signed webhooks	Checkout sessions; events per FR-16.02	Idempotent handlers by event id; reconciliation job daily
SendGrid	Transactional email	API key	Dynamic templates send; bounce/complaint webhooks	Retry + suppression list (FR-18.03)
Economic calendar feed (Q7)	News filter	Vendor key	High-impact events pull (15-min sync)	Stale feed >60 min → news filter fails open with warning banner (entry risk accepted, logged)

Integration	Purpose	Auth	Key operations / events consumed	Failure handling
Market data (Dukascopy, Binance dumps)	Historical OHLCV	Public/licensed	Nightly ingestion, gap backfill	Checksum + dataset versioning (FR-11.01)
PostHog	Product analytics	Project key	Event taxonomy (§7 / PRD §13)	Client + server capture; no PII in properties
Sentry / Prometheus	Errors & metrics	Keys	Traces w/ correlation ids (webhook→order)	n/a
Cloudflare	WAF, rate limits on hook ingress	—	IP throttles, bot rules	Managed rules + custom per FR- 09.03

5. Data Requirements (dictionary — core tables, abridged)

Conventions: `(id uuid v7 pk)` · `(tenant_id uuid not null)` (RLS) · `(created_at/updated_at timestamptz)` · money `(bigint)` minor units + `(currency char(3))`

- soft-delete only where stated.

Table	Key columns (type — constraint)
users	email citext unique(tenant), password_hash, display_name, country char(2), tz text, flags jsonb, twofa_enabled bool
consents	user_id fk, doc_type enum(tos,privacy,risk,auto_consent), version, ip inet, ts
broker_connections	user_id fk, provider enum(metaapi,ccxt_binance,ccxt_bybit,ctrader), status enum(connecting,synchronizing,live,degraded,disconnected), encrypted_secret_ref text, meta jsonb, unique(user_id, nickname)
accounts	connection_id fk null(paper), kind enum(live,paper), base_ccy, balance_minor, equity_minor, leverage int, reset_generation int default 0
positions	account_id fk, provider_position_id text unique(connection), symbol, side, volume numeric, entry_price numeric, sl, tp, opened_at, source enum
orders	account_id fk, provider_order_id text unique(connection), state enum per FR-07.03 (check constraint on transitions), symbol, side, type, volume, price, sl, tp, tif, source enum(manual,signal,webhook,external,system), idempotency_key text unique(account), reason_code text
executions	order_id fk, provider_deal_id text unique(connection), fill_price, fill_volume, fee_minor, ts
sync_anomalies	account_id, type enum, entity, before jsonb, after jsonb
signals	strategy definitions per FR-08.01 (versioned)
signal_instances	user_id, strategy_id+version, config jsonb (schema-validated), mode enum(notify,confirm,auto), status, target_account fk
webhook_endpoints	user_id, slug uuid unique, secret_ref, target_account fk, template jsonb, symbol_map jsonb, status
webhook_events	endpoint_id, raw jsonb (secrets masked), decision enum(executed,blocked,duplicate,error), reason_code, latency_ms, ts (30/90d retention)
guards	account_id, type enum per FR-10.04, params jsonb, status, challenge_tag text null

Table	Key columns (type — constraint)
guard_events	guard_id, order_intent_ref, outcome enum(blocked,transformed,triggered), code, ts
backtests	user_id, config_snapshot jsonb, dataset_version fk, status, progress, results jsonb, expires_at
courses/modules/lessons/quiz_attempts/progress	per FR-03.* (lessons versioned)
points_tx	user_id, type enum(earn,redeem,expire,adjust), amount int, reason_code, ref, ts (append-only)
referral_edges	referrer_id, referee_id unique, level smallint check(1..2 default), created_at
referral_rewards	edge_id, status enum(hold,issued,clawed_back), reward jsonb, hold_until
affiliates / affiliate_deals	partner profile; deals versioned jsonb per FR-15.02, effective_from
attributions	user_id unique (one owner), owner_type enum(referral,affiliate), owner_id, locked_at
ledger_entries	journal_id, account text, direction enum(debit,credit), amount_minor, currency, ref jsonb — invariant $\Sigma \text{debit} = \Sigma \text{credit}$ per journal (deferred constraint + nightly check)
payouts	affiliate_id, amount_minor, status enum per FR-15.05, approvals jsonb, external_ref
share_cards	user_id, type, payload jsonb, image_url, disclaimer_version
notifications / preferences	per FRD-18
audit_log	actor_id, role, action, entity, before jsonb, after jsonb, ip, ts (append-only, no delete)
tenant_config_versions	tenant_id, version, config jsonb, published_by, published_at
Timescale: ohlcv, calendar_events, dataset_versions	per FR-11.01

Retention: trades/executions ≥7y (E5.7) · webhook_events 30d (Elite 90d) · audit_log ≥7y · backtest results 90d/1y · GDPR delete anonymizes PII, preserves financial + audit records under legal basis.

6. Reporting Requirements

Business (weekly beta / monthly GA): activation funnel, free→paid, MRR & churn (Stripe recon), IB rebate revenue (ledger), K-factor, guard adoption & guarded-vs-unguarded churn delta, north-star (Weekly Guarded Live Traders). Ops (real-time): FR-17.07 dashboards. Partner: FR-15.06. Finance (monthly): revenue by stream, MetaAPI cost per account, ledger trial balance, payout register, LTD seats.

7. Instrumentation Reference

Event taxonomy = PRD §13 (source of truth); server-side emission for all money/execution events; client-side for navigation/paywall; correlation id propagated
webhook→intent→order→notification; no PII in event properties (user id only).

8. Non-Functional Requirements (binding reference — PRD §10)

Key figures restated for QA: webhook→submit p50 ≤1.5s / p95 ≤3s (excl. venue) · signal bar-close→submit p95 ≤5s · stream→UI ≤3s · availability 99.5% beta / 99.9% GA on execution path · live path fails closed (FR-10.03) · OWASP ASVS L2, envelope encryption, secrets never logged (CI scrubber test), 2FA before live connect, full audit trail · RPO ≤15 min, RTO ≤4h · pen test before GA.

9. Assumptions, Dependencies & Open Items

Assumptions: MetaAPI supports required server directory + probe operations; Binance/Bybit expose key-permission checks (else fallback: attempt-withdrawal-scope canary is NOT used — manual attestation + docs); TradingView cannot sign headers (secret-in-payload fallback stands); delayed paper feed acceptable at v1 (Q9).
Dependencies & open items inherit PRD §17 (Q1-Q10); FRD-blocking subset: Q3 (PSP) blocks FRD-16 go-live at M3 · Q4 (AI-tips language) blocks FR-08.07 copy at Closed Beta · Q5 (country list) blocks FR-17.03 defaults at M1 · Q7 (calendar feed) blocks FR-10.04 news filter (P1) · Q9 (paper feed) blocks FR-04.02 final config at M0.

Appendix A — Error Catalog (excerpt; stable codes)

error_code	HTTP	User copy (EN)	Remediation
2fa_required	403	Two-factor authentication is required before connecting a live account.	Inline 2FA setup, resume flow
upgrade_required	402	Live connections aren't available on your plan.	Plan sheet
investor_password	422	That looks like an investor (read-only) password.	Use the trading password
key_scope_withdrawal	422	This API key allows withdrawals — for your safety we can't accept it.	Key-creation guide (read+trade only)
unmapped_symbol	422	We don't know how to map {ticker} to your broker.	One-click add-mapping
budget_exceeded	429	Monthly alert budget reached.	Upgrade / resets {date}
blocked_by_guard:daily_loss	200*	Daily-loss guard blocked this entry.	Guard settings link (*webhook semantics FR-09.07)
blocked_by_guard:session_filter	200*	Outside your trading session window.	Edit windows
system_guard_unavailable	200*/503	Automation paused — safety first. We couldn't verify your guards.	Auto-retry info, status page
risk_too_small_for_min_lot	422	Your risk % gives a size below the broker minimum.	Raise risk %, widen SL, or smaller min-lot symbol
insufficient_margin	422	Not enough margin for this size.	Reduce size
invalid_stops_distance	422	SL/TP too close to price (min {X} points).	Adjust levels
market_closed	422	Market is closed for {symbol}.	Session hours link
compliance_signoff_required	403	This referral configuration requires Compliance sign-	BRL-03 notice

error_code	HTTP	User copy (EN)	Remediation
		off.	
ltd_sold_out	410	The Lifetime Deal is sold out.	Waitlist
duplicate	200	(webhook) Duplicate alert ignored.	—

Appendix B — Webhook Response Contract

`POST /wh/{uuid}` → 200 `{status: executed|blocked_by_guard|duplicate|budget_exceeded*|error, order_id?, reason_code?, latency_ms}` (*429 for budget) · 401 bad signature · 410 endpoint disabled · 422 validation with reason_code · Response always ≤2KB JSON; no HTML.

Appendix C — Consent Records (compliance-critical)

auto_consent (full-auto enablement, FR-08.05): {user, account, instance, text_version, ts, ip, expires_at = ts+90d}; expiry job downgrades mode auto→confirm with notification. All consent texts versioned in tenant config; re-acceptance required on material change (PRD C8).

Appendix D — Traceability to Test Plan

QA shall derive at minimum: idempotency suite (FR-09.04, FR-08.04), fail-closed chaos suite (FR-10.03: kill guard service under live webhook load), ledger balance property tests (FR-15.04), entitlement bypass attempts (FR-00.01, FR-16.03), LTD concurrency (FR-16.05), RLS isolation (FR-20.01), secret-leak log scan (FR-05.06), determinism (FR-11.04), referral compliance gate (FR-14.03). Milestone gates M1/M5 (PRD §15) require these suites green.

— End of FRD v1.0 —